Question	Answer	Marks	Partial Marks
5(b)	24.9 or 24.94 to 24.95	**5**	**B1** for angle $PQS = 42$ **M2** for $QS = 5.9 \div \cos 42$ oe or **M1** for $\cos 42 = \frac{5.9}{QS}$ oe **M1**dep for *their* $SQ \times \pi$ oe
6(a)(i)	9.5, 4.8 and 8.5	**3**	**B1** for each
6(a)(ii)	correct curve	**5**	**B4** for correct curve, but branches joined or touching y axis or **B3FT** for 9 or 10 correct plots or **B2FT** for 7 or 8 correct plots or **B1FT** for 5 or 6 correct plots AND **B1 indep** two separate branches not touching or cutting y-axis
6(b)	$y = \frac{24}{5} - 2x$ ruled and – 0.4 to – 0.2 and 1.45 to 1.7	**4**	**B2** for correct ruled line crossing curve twice or **B1** for correct freehand or for short ruled line or for line with negative gradient through (0, 4.8) or for line with gradient – 2 **B1** for each value
6(c)	[a =] 10 [b =] 20 [c =] – 48	**4**	**B3** for $10x^3 - 15 = 48x - 20x^2$ oe or better or **B2** for 2 correct values or **B1** for 1 correct value or for $5x^2 - \frac{15}{2x} = 24 - 10x$ or better or for $2x^3 - 3 = \frac{48}{5}x - 4x^2$ or better or for $x^3 - \frac{3}{2} = \frac{24}{5}x - 2x^2$ After 0 scored SC1 for correct elimination of a denominator of 5, x or $2x$ from a four term expression.
7(a)(i)(a)	Shape at (–2, 1) (–4, 1) (–4, 7) (0, 7)	**2**	**B1** for 3 correct points or for enlargement SF2 from any centre
7(a)(i)(b)	Shape at (2, –2) (2, –3) (5, –1) (5, –3)	**3**	**B2** for correct orientation but wrong position or for 3 correct points or **B1** for $y = x - 1$ drawn

Question	Answer	Marks	Partial Marks
7(a)(ii)	Rotation 90 [anticlockwise] oe (0, 0) oe	**3**	**B1** for each
7(b)	$\frac{3}{4}\mathbf{p}+\frac{1}{2}\mathbf{q}$ or $\frac{1}{4}(3\mathbf{p}+2\mathbf{q})$ or $\frac{3\boldsymbol{p}+2\boldsymbol{q}}{4}$ final answer	**3**	**M2** for $AM = \overrightarrow{AM} = \frac{1}{2}\left(-\mathbf{p}+\mathbf{q}+\frac{1}{2}\mathbf{p}\right)$ oe or **M1** for correct route for $\overrightarrow{AB}$ oe soi by –½ **p** + **q** or for $\overrightarrow{OM}$ soi
8(a)(i)	1.6 oe	**2**	**M1** for $3 - 5x = -5$
8(a)(ii)	$\frac{3-x}{5}$ oe final answer	**2**	**M1** for $x = 3 - 5y$ or $\frac{y}{5}=\frac{3}{5}-x$ or better, or $y - 3 = -5x$ oe
8(b)(i)	$20.25-(1.5+x)^2$	**3**	Method 1 **B1** for $(\pm 1.5 \pm x)^2$ seen **B1** for [b =] 18 + *their* 1.5^2 OR Method 2 **B1** for $b - a^2 - 2ax - x^2$ or for b = 20.25 **B1** for a = 1.5
8(b)(ii)	Correct sketch with max in correct quadrant at $(-1.5, 20.25)$	**3**	**FT** *their* $20.25-(\textit{their}1.5+x)^2$ provided in that form **B1** for $\cap$ shape or for $\cup$ shape if in form $c+(d+x)^2$ in part **(b)(i)** **B1** for TP at $(-1.5, k)$ or $(k, 20.25)$ FT *their* $20.25 \pm (\textit{their}1.5+x)^2$ or for (–1.5, 20.25) seen
8(b)(iii)	[y =] $34 - 11x$	**6**	**B2** for $-3 - 2x$ or **B1** for either kx –3, $k \neq 0$ or $-2x$ + n or for $18 - 3 - 2x$ **M1dep** for gradient = *their* (–3 – 2(4)) **B1** for y-value at x = 4, is –10 **M1dep** for *their* –10 = (*their* –11)4 + c oe